

Final Project: Data Ethics and Policy

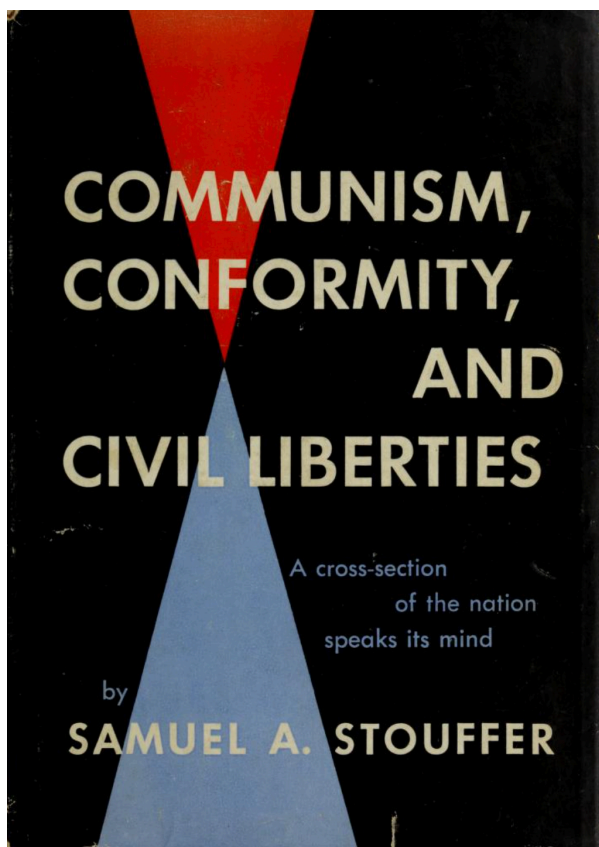
Christy Hsu

Abstract

This project looked into an attempt in the fifties that set out to measure individual tolerance and use that measure to evaluate the impact the anti-Communism Cold War agenda—both abroad and at home—had on U.S. citizens. I saw the picking up of this 70-year-old survey data, and the analysis that accompanied it, as a chance to stroll through the data ethics concerns that can arise when holding the hope to learn about the world from data

Introduction

The online poll data library host by Roper Center was where I first came across the Stouffer Study of 1954 *Communism, Conformity, and Civil Liberties: A Cross-Section of the Nation Speaks Its Mind* by Samuel Andrew Stouffer([Fund For The Republic and Stouffer 2020](#))And, I gained access to the entire dataset on ICPSR([Samuel A. Stouffer 1992](#))



I found Stouffer's attempt in the fifties to design public opinion polls and, construct an innovative way of measuring the latent properties tolerance and fear at the individual level very interesting. The tolerance scale and the perception of internal communist danger scale are not included in the data, thus a major part of this project involved returning these two target variables in order to complete the picture and reproduce that basis on which Stouffer built his arguments. I learned much from this practice of reading and researching that past effort into conceptualizing and operationalizing tolerance and fear, and got my hands dirty to really apply those framings and methods to the data. It gave me a chance to reflect on the many artistic and arbitrary decisions that the researcher made throughout this data analysis process.

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Harvesting from Historical Data Collection Effort: A More Friendly Format

Complying ICPSR's redistribution policy, the converted data files are not provided here. Instead, the author provides STATA .do and .dct files, which were constructed based on the

reading of the codebook. Please download the dataset in ASCII format from ICPSR and should be able to apply to decode the .txt files from both samples. (Samuel A. Stouffer 1992)

```
read-ascii-files
  gp-decode.do
  lead-decode.do
  sample1.dct
  sample2.dct
```

Return target variables

The tolerance that Stouffer argued upon. (Samuel Andrew Stouffer 1955, 262–69)

Prepare the code_df data frame

1. Cleaning column names and binding the two samples

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Adding Binary and Ternary Variables leader, interested and categorizer: these are the variables that Stouffer specified in his book as ways to divide the respondents and make comparisons.

leader	n
0	4933
1	1500

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leader	interested	n	pct
0	less	2155	43.68538
0	more	2778	56.31462
1	less	210	14.00000
1	more	1290	86.00000

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Source: [Article Notebook](#)

leader	categorizer	n	pct
0	agree	3127	63.389418
0	disagree	1495	30.306102
0	dont know	311	6.304480
1	agree	806	53.733333
1	disagree	669	44.600000
1	dont know	25	1.666667

Source: [Article Notebook](#)

Scale1: Willingness to Tolerate Nonconformist

Conceptual Tolerance and Operational Tolerance

The questionnaires used to rank respondents into six tolerance groups focused on four types of nonconformists:

- A person who is against all churches and religion (atheist)
- A person who favors government ownership (socialist)
- An alleged communist (someone whose loyalty has been questioned by a Congressional Committee but swears under oath they have never been a communist)
- An admitted communist

respondent were asked about their approval of 3 types of disposition against the above nonconformist, and whether they agree or disagree the limitation or deprivation of the nonconformist's civil liberties, for example:

1. Freedom speech:

- "If ____ wants to make a speech in your community, should he be allowed to speak or not?"

2. Book censor:

- "Suppose he wrote a book that is in your public library. Somebody in your community suggests the book should be removed. Would you favor removing it, or not?"

3. Employment:

- Should a radio singer who is a nonconformist be fired or not?

- Should a college or university teacher be fired or not?
- Should a high school teacher be fired or not?
- Should someone working in a defense plant be fired or not?
- Should a store clerk be fired or not?

4. Boycott:

- “Suppose the radio program he is on advertises a brand of soap. Somebody in your community suggests you stop buying that soap. Would you stop or not?”

0 to 5: Scaling Individual Tolerance

I ran into many challenges replicating Stouffer’s results. Both the overall proportions across tolerance rankings, was unable to reproduce the group counts applying further breakdowns, such as by age, region, education, thus for comparison.

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leader	tolerance_group	n	pct
0	tolerance0	651	13.196838
0	tolerance1	871	17.656598
0	tolerance2	1263	25.603081
0	tolerance3	951	19.278330
0	tolerance4	395	8.007298
0	tolerance5	802	16.257855
1	tolerance0	79	5.266667
1	tolerance1	182	12.133333
1	tolerance2	180	12.000000
1	tolerance3	246	16.400000
1	tolerance4	205	13.666667
1	tolerance5	608	40.533333

Source: [Article Notebook](#)

tolerance_group	n
tolerance0	730
tolerance1	1053
tolerance2	1443
tolerance3	1197
tolerance4	600
tolerance5	1410

Source: [Article Notebook](#)

Broader Tolerance Rank Groups: less tolerant, in-between and more tolerant

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tolerance_broader0	n	pct
more tolerant	813	54.2
in between	426	28.4
less tolerant	261	17.4

Source: [Article Notebook](#)

Attempt2

Allowing some inconsistency?

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tolerance	n
tolerance0	674
tolerance1	530
tolerance2	1334
tolerance3	1229
tolerance4	807
tolerance5	1859

Source: [Article Notebook](#)

leader	tolerance	n	pct
0	tolerance0	604	12.244070
0	tolerance1	491	9.953375
0	tolerance2	1179	23.900264
0	tolerance3	993	20.129738
0	tolerance4	554	11.230488
0	tolerance5	1112	22.542064
1	tolerance0	70	4.666667
1	tolerance1	39	2.600000
1	tolerance2	155	10.333333
1	tolerance3	236	15.733333
1	tolerance4	253	16.866667
1	tolerance5	747	49.800000

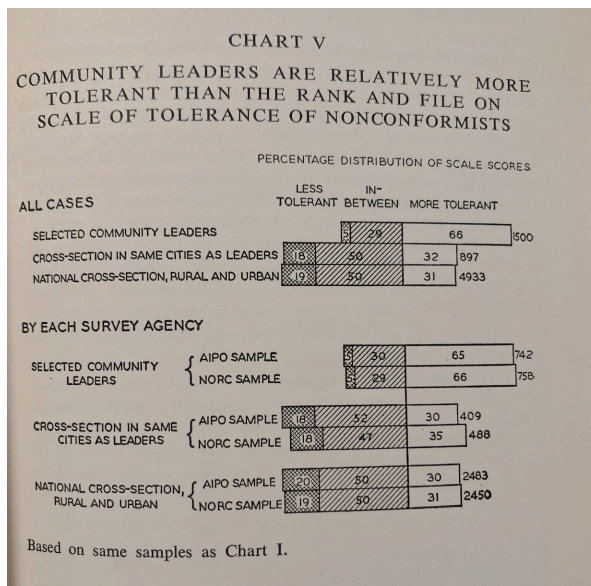
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leader	tolerance__broader	n	pct
0	more tolerant	1666	33.772552
0	in between	2172	44.030002
0	less tolerant	1095	22.197446
1	more tolerant	1000	66.666667
1	in between	391	26.066667
1	less tolerant	109	7.266667

Source: [Article Notebook](#)

To answer this question([Samuel Andrew Stouffer 1955, 51](#))



us region

Source: [Article Notebook](#)

us_region	tolerance_broader	n	pct
east	more tolerant	276	68.148148
east	in between	107	26.419753
east	less tolerant	22	5.432099
midwest	more tolerant	348	69.322709
midwest	in between	123	24.501992
midwest	less tolerant	31	6.175299
south	more tolerant	225	57.251908
south	in between	123	31.297710
south	less tolerant	45	11.450382
west	more tolerant	151	75.500000
west	in between	38	19.000000
west	less tolerant	11	5.500000

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Scale2: Scale of the Perception on the Internal Communist Danger

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Source: [Article Notebook](#)

danger	n
danger0	507
danger1	682
danger2	2116
danger3	1170
danger4	1093
danger5	865

Source: [Article Notebook](#)

Broader rank groups

Source: [Article Notebook](#)

danger_broader	n
great threat	1958
in between	3286
little threat	1189

Source: [Article Notebook](#)

Source: [Article Notebook](#)

Evaluating Operationalizations: Reliability and Validity, insights from classification algorithms

Data and Measures: Validity

From the conceptual variable tolerance to the operationalized definition of tolerance, Stouffer proposed *h-technique* to map answers of the respondent to a tolerance score corresponding to their degree of tolerance.([STOUFFER et al. 1952](#)) But does this tolerance scale really measuring people's tolerance or is it measuring something else?([Tessler 2022, 43–47](#))

Reliability

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3-class classification using the strict measure of tolerance score tolerance_broader0

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tolerance_broader0	n
in between	426
less tolerant	261
more tolerant	813

Source: [Article Notebook](#)

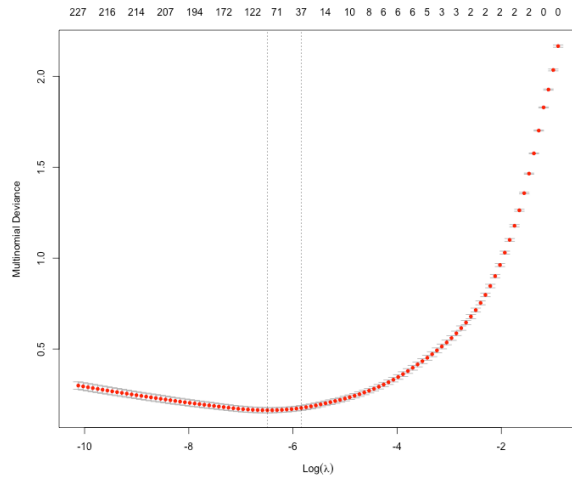
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```
clf0 <- cv.glmnet(  
  x_train, y_train, family = "multinomial",  
  alpha = 1, type.multinomial = "ungrouped"  
)  
  
lambda_1se0 <- clf0$lambda.1se  
lambda_1se0  
# 0.002905948  
  
png("clf0-plot.png", width = 700, height = 600)  
plot(clf0)  
dev.off()
```



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[1] 0.9902837

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[1] 0.989899

Source: [Article Notebook](#)

Confusion Matrix and Statistics

	Reference		
Prediction	more tolerant	in between	less tolerant
more tolerant	400	4	3
in between	0	511	3
less tolerant	1	2	363

Overall Statistics

Accuracy : 0.9899
 95% CI : (0.9828, 0.9946)
 No Information Rate : 0.4017
 P-Value [Acc > NIR] : <2e-16

Kappa : 0.9847

McNemar's Test P-Value : 0.1577

Statistics by Class:

	Class: more tolerant	Class: in between
Sensitivity	0.9975	0.9884
Specificity	0.9921	0.9961
Pos Pred Value	0.9828	0.9942
Neg Pred Value	0.9989	0.9922
Prevalence	0.3116	0.4017
Detection Rate	0.3108	0.3970
Detection Prevalence	0.3162	0.3994
Balanced Accuracy	0.9948	0.9922
	Class: less tolerant	
Sensitivity	0.9837	
Specificity	0.9967	
Pos Pred Value	0.9918	
Neg Pred Value	0.9935	
Prevalence	0.2867	
Detection Rate	0.2821	
Detection Prevalence	0.2844	
Balanced Accuracy	0.9902	

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predictor	n
(Intercept)	3
admitted_communist_allow_speak5	2
admitted_communist_buy_soap5	2
admitted_communist_put_jail5	2
alledged_communist_fire_teacher5	2
censor_antireligion_book5	2
censor_antireligion_speaker5	2
censor_antireligion_speaker8	2
censor_socialist_book5	2
censor_socialist_speaker8	2
many_communists_defense_plants8	2
many_communists_us5	2

predictor	n
-----------	---

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3-class classification: tolerance_broader based on the measure with wiggle room

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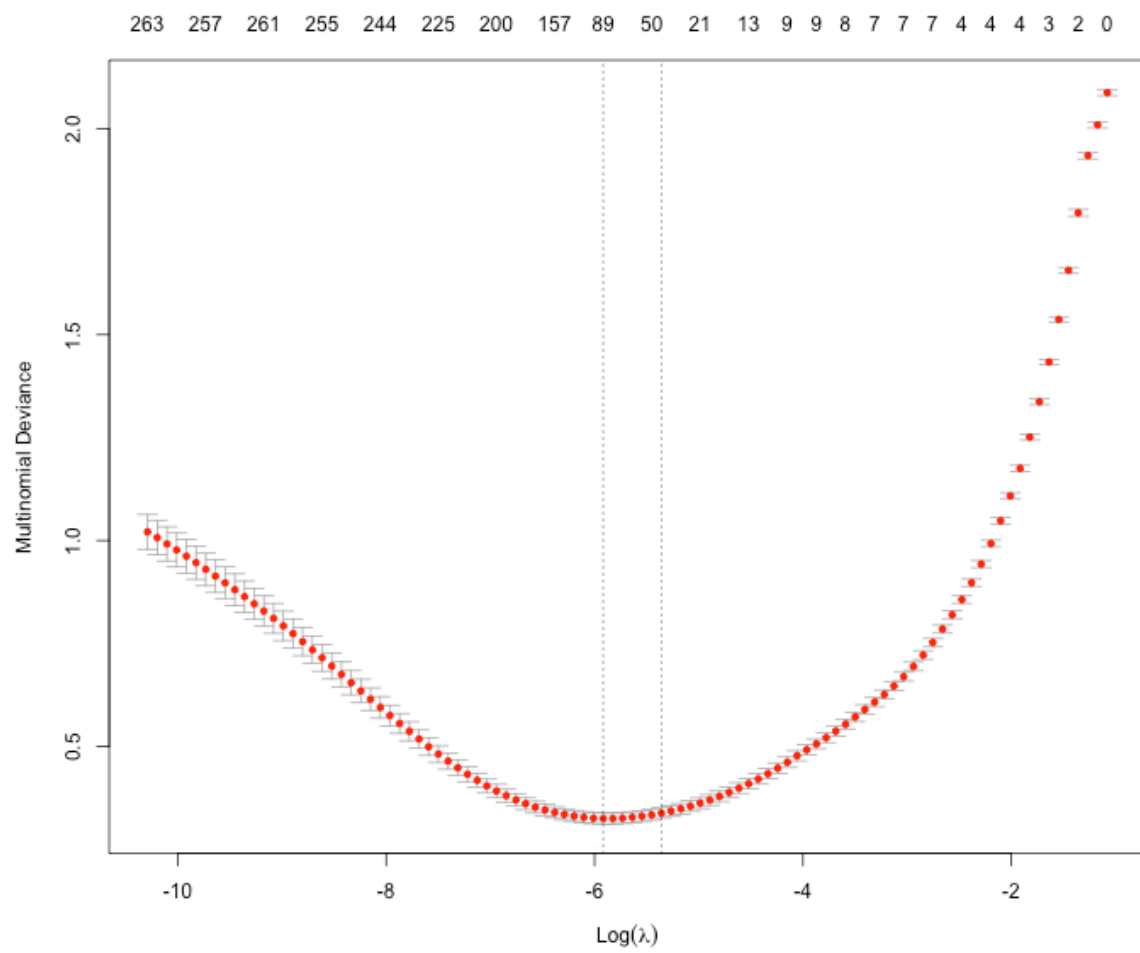
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Source: [Article Notebook](#)

Source: [Article Notebook](#)

```
clf <- cv.glmnet(  
  x_train, y_train, family = "multinomial",  
  alpha = 1, type.multinomial = "ungrouped"  
)  
# png("clf-plot.png", width = 700, height = 600)  
# plot(clf)  
# dev.off()  
lambda_best <- clf$lambda.1se  
# 0.00391  
lambda_seq <- clf$lambda  
# saveRDS(lambda_seq, file = "data/clf-lambda-seq.rds")
```



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[1] 0.958803

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[1] 0.950272

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Confusion Matrix and Statistics

	Reference		
Prediction	more tolerant	in between	less tolerant
more tolerant	505	4	5
in between	9	489	12
less tolerant	7	27	229

Overall Statistics

Accuracy : 0.9503
 95% CI : (0.9369, 0.9615)
 No Information Rate : 0.4048
 P-Value [Acc > NIR] : < 2e-16

Kappa : 0.9222

McNemar's Test P-Value : 0.04548

Statistics by Class:

	Class: more tolerant	Class: in between
Sensitivity	0.9693	0.9404
Specificity	0.9883	0.9726
Pos Pred Value	0.9825	0.9588
Neg Pred Value	0.9793	0.9601
Prevalence	0.4048	0.4040
Detection Rate	0.3924	0.3800
Detection Prevalence	0.3994	0.3963
Balanced Accuracy	0.9788	0.9565
	Class: less tolerant	
Sensitivity	0.9309	
Specificity	0.9673	
Pos Pred Value	0.8707	
Neg Pred Value	0.9834	
Prevalence	0.1911	
Detection Rate	0.1779	
Detection Prevalence	0.2044	
Balanced Accuracy	0.9491	

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Source: [Article Notebook](#)

predictor	n
(Intercept)	3
admitted_communist_allow_speak5	2
admitted_communist_buy_soap5	2
admitted_communist_put_jail5	2
admitted_communist_remove_book_library5	2
alleged_communist_fire_clerk5	2
alleged_communist_fire_clerk8	2
alleged_communist_fire_defense_plant5	2
alleged_communist_fire_high_school_teacher5	2
alleged_communist_fire_singer5	2
alleged_communist_fire_teacher5	2
alleged_communist_make_community_speech5	2

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Are these selected Items capturing most of the variances?

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Learning: Predicting Tolerance Score without the original 15 items

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Source: [Article Notebook](#)

Source: [Article Notebook](#)

tolerance_broader0	n
in between	426
less tolerant	261
more tolerant	813

Source: [Article Notebook](#)

Source: [Article Notebook](#)

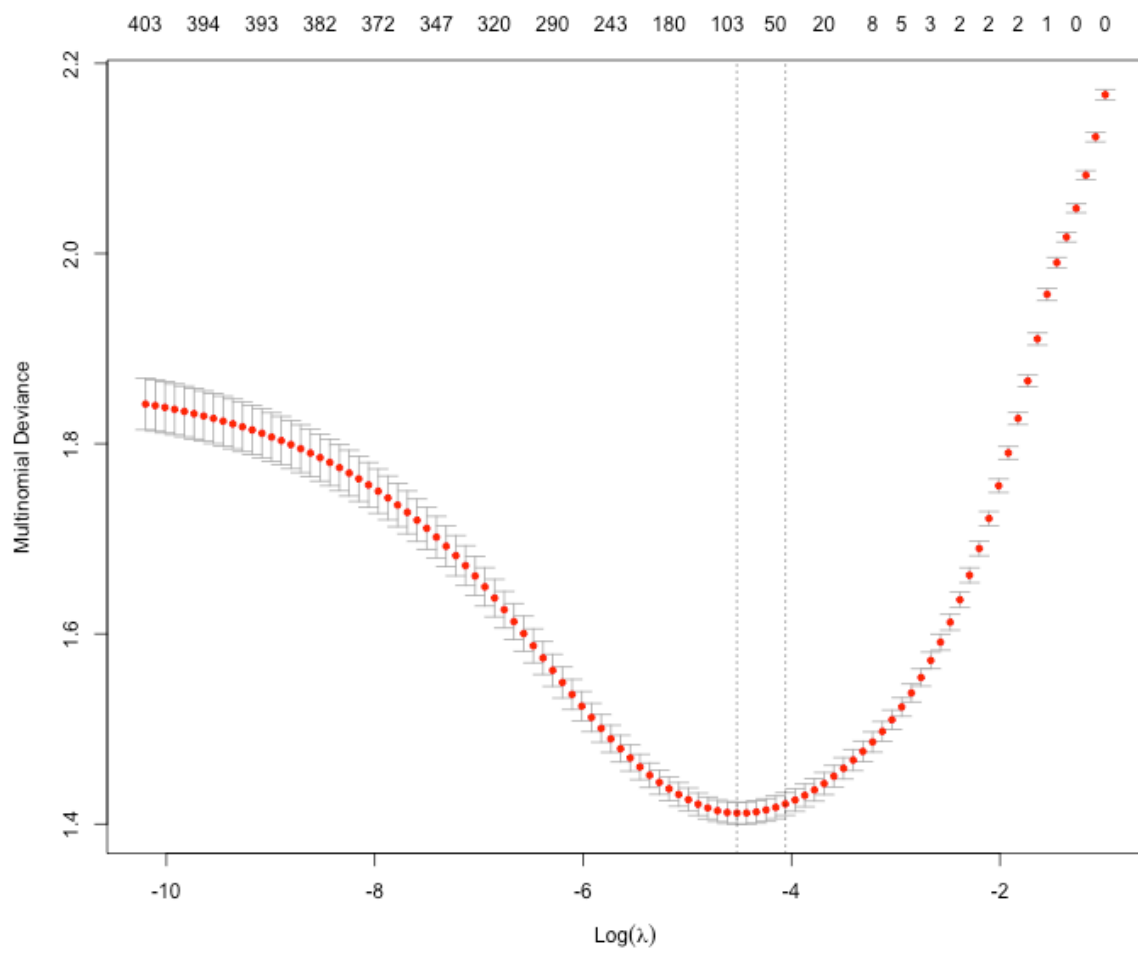
Source: [Article Notebook](#)

Source: [Article Notebook](#)

Source: [Article Notebook](#)

Source: [Article Notebook](#)

```
clf3 <- cv.glmnet(  
  x_train, y_train, family = "multinomial",  
  alpha = 0.75, type.multinomial = "ungrouped"  
)  
lambda_1se3 <- clf3$lambda.1se  
lambda_1se3  
# 0.01726083  
lambda_seq3 <- clf3$lambda  
# saveRDS(lambda_seq3, file = "data/clf3-lambda-seq.rds")  
  
# png("image/clf3-plot.png", width = 700, height = 600)  
# plot(clf3)  
# dev.off()
```



Source: [Article Notebook](#)

[1] 0.7188107

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[1] 0.6985237

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Confusion Matrix and Statistics

Prediction	Reference		
	more tolerant	in between	less tolerant
more tolerant	264	65	39
in between	125	435	130
less tolerant	12	17	200

Overall Statistics

Accuracy : 0.6985
 95% CI : (0.6726, 0.7235)
 No Information Rate : 0.4017
 P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.5322

McNemar's Test P-Value : < 2.2e-16

Statistics by Class:

	Class: more tolerant	Class: in between
Sensitivity	0.6584	0.8414
Specificity	0.8826	0.6688
Pos Pred Value	0.7174	0.6304
Neg Pred Value	0.8509	0.8626
Prevalence	0.3116	0.4017
Detection Rate	0.2051	0.3380
Detection Prevalence	0.2859	0.5361
Balanced Accuracy	0.7705	0.7551
	Class: less tolerant	
Sensitivity	0.5420	
Specificity	0.9684	
Pos Pred Value	0.8734	
Neg Pred Value	0.8403	
Prevalence	0.2867	
Detection Rate	0.1554	
Detection Prevalence	0.1779	
Balanced Accuracy	0.7552	

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Acquired the class specific variables and their coefficients

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predictor	n
(Intercept)	3
admitted_communist_fire_clerk5	2
admitted_communist_lose_citizenship5	2
break_friendship_former_communist5	2
censor_antireligion_teacher5	2
child_rearing_respect3	2
last_grade_finished_school5	2
remember_hiss_caught8	2
admitted_communist_fire_college_professor5	1
admitted_communist_lose_citizenship8	1
alleged_communist_buy_soap5	1
alleged_communist_fire_singer5	1

Source: [Article Notebook](#)

Worrying about Internal Communist Threats

Pearson's Chi-squared test

```
data:  toler_rigid_tb
X-squared = 491.09, df = 10, p-value < 2.2e-16
```

Source: [Article Notebook](#)

```
      agree  disagree  dont know
tolerance0  3.6776281 -5.3964437  1.1128421
tolerance1 -2.3955483 -1.0211719 10.7874512
tolerance2  4.5040579 -6.1915000  0.3030713
tolerance3  2.7420911 -2.5744267 -2.8481376
tolerance4 -0.6175252  2.2641002 -3.6331099
tolerance5 -7.2561266 11.9240239 -5.4354457
```

Source: [Article Notebook](#)

Pearson's Chi-squared test

```
data:  toler_danger_tb
X-squared = 410.67, df = 25, p-value < 2.2e-16
```

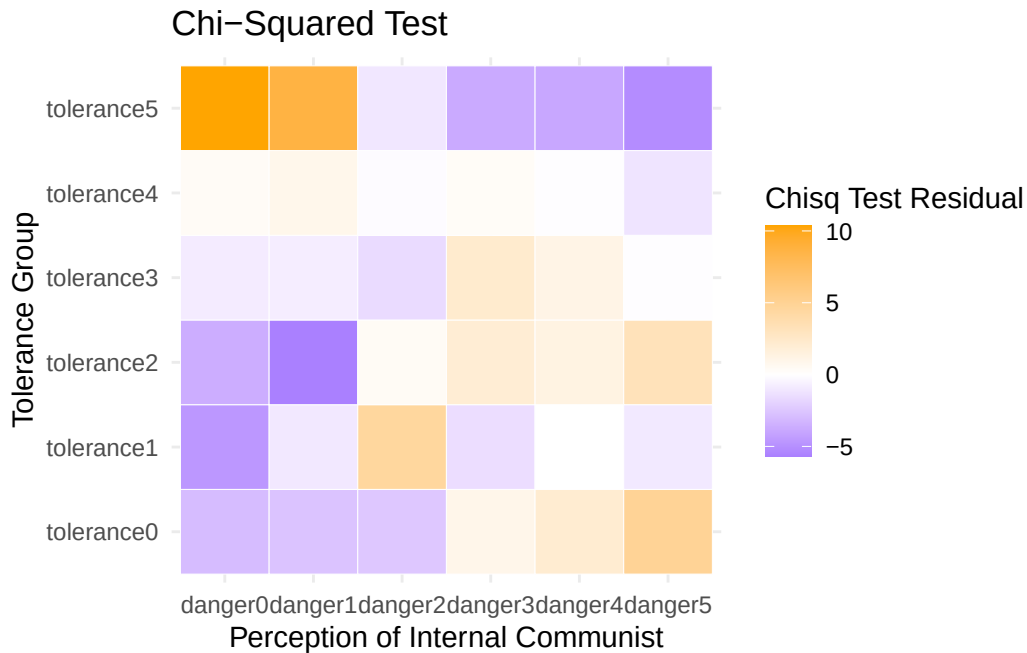
Source: [Article Notebook](#)

	danger0	danger1	danger2	danger3	danger4
tolerance0	-2.970715176	-2.658964979	-2.459910036	0.974739884	2.152232514
tolerance1	-4.609232171	-1.006527133	4.547776892	-1.482352369	0.006717332
tolerance2	-3.631408736	-5.657972356	0.475321324	2.009535358	1.330107400
tolerance3	-0.858497500	-0.790142288	-1.598983893	2.256620306	1.165660786
tolerance4	0.394465523	0.926642029	-0.167803951	0.370962087	-0.093407449
tolerance5	10.328080755	8.630383914	-1.058231155	-3.774441827	-3.913075010

	danger5
tolerance0	4.929822736
tolerance1	-0.973974888
tolerance2	3.228408105
tolerance3	-0.075048875
tolerance4	-1.188784303
tolerance5	-5.126832308

Source: [Article Notebook](#)

Source: [Article Notebook](#)



Source: [Article Notebook](#)

Results

Reflections

Policy Recommendation

Fund For The Republic, and Samuel A. Stouffer. 2020. "Communism, Conformity & Civil Liberties." Roper Center for Public Opinion Research. <https://doi.org/10.25940/ROPER-31089514>.

Stouffer, Samuel A. 1992. "Communism, Conformity, and Civil Liberties Study, 1954." Inter-university Consortium for Political and Social Research. <https://doi.org/10.3886/ICPSR07202.v2>.

STOUFFER, SAMUEL A., EDGAR F. BORGATTA, DAVID G. HAYS, and ANDREW F. HENRY. 1952. "A Technique for Improving Cumulative Scales*." *Public Opinion Quarterly* 16 (2): 273–91. <https://doi.org/10.1086/266388>.

Stouffer, Samuel Andrew. 1955. *Communism, Conformity, and Civil Liberties: A Cross-Section of the Nation Speaks Its Mind*. Transaction Publishers.

Tessler, Mark. 2022. *Social Science Research in the Arab World and Beyond: A Guide for Students, Instructors and Researchers*. Springer Nature.